1. SIGNAL GENERATION AND BASIC SIGNAL OPERATIONS

Experiment 1.2

Generation of Discrete Time Signals

Aim:

To generate and verify a discrete-time unit impulse signal using MATLAB. The Aim is to ensure that the signal generator can accurately produce an impulse signal at a specified time index, which is crucial for analyzing and understanding the behavior of discrete-time systems.

Apparatus Required

PC with MATLAB.

Program:

```
% Generation of Discrete Time Signal

clc;

clear;

close all;

% Discrete-time index

n = -10:10;

% Discrete-time sinusoidal signal

x = sin(0.2*pi*n);

% Plotting

figure;

stem(n, x, 'filled');

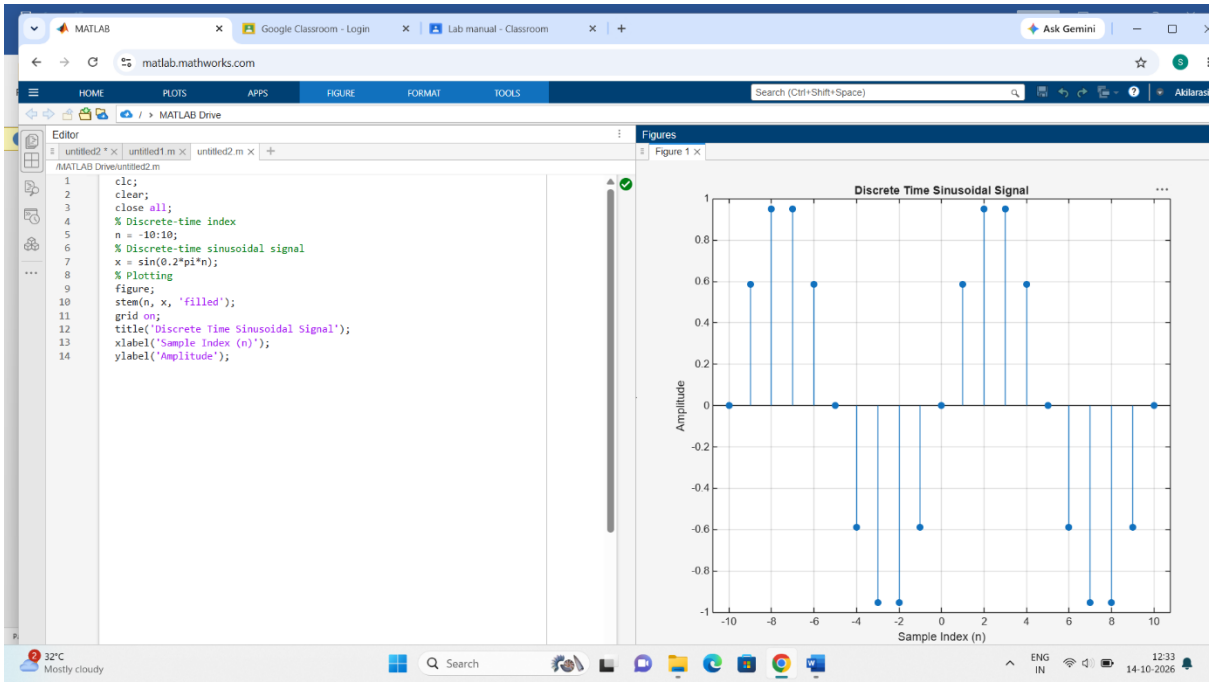
grid on;

title('Discrete Time Sinusoidal Signal');

xlabel('Sample Index (n)');

ylabel('Amplitude');
```

Output Waveform:



Result:

This test case validates that the signal generator can accurately produce a discrete-time unit impulse signal at a specified index, which is essential for analyzing the response of discrete-time systems.